

Chapter – 3

SINTER PLANT

3.1 Introduction

Sinter Plant agglomerates iron ore fines with other fine materials at high temperature, such that constituent materials fuse together to make a single porous mass.

A large quantity of iron ore fines is generated in the mines, which cannot be charged directly into the Blast furnace. Moreover many metallurgical wastes are generated in the steel industry itself, disposal of which is very difficult. In order to consume this otherwise waste fine materials, they are agglomerated together and made into lumps by a process known as SINTERING.

Sintering is the process of agglomeration of fines (steel plant wastes) by incipient fusion caused by heat available from the fuel contained in the charge. This technology was developed for the treatment of waste fines in the early 20th century. Since then sinter has become the widely accepted & preferred Blast furnace burden material.

Raw materials used in Sinter Plant

1. Iron ore fines
2. Lime stone fines
 - Dolomite fines
 - Coke breeze fines
 - B.O.F.Sludge
 - Burnt Lime
 - Mills Scale
 - B.O.F.Slag / L D slag
 - BF Return fines
 - Internal Sinter Return fines

3.2 Sintering Process

The Iron ore fines, lime stone fines, dolomite fines, lime dust, coke breeze and other metallurgical wastes are proportioned based on charge calculation. These charge thus mixed in a balling drum with the addition of water and then loaded into grates of moving pallets. The purpose of Balling drum is to mix the raw materials (called base mix) with water and make balls. After mixing and ball formation (nodulization

) this base mix (now called green mix) is loaded on moving sinter machine pallets. HEARTH LAYER which consists of finished sinter of size fraction 10 to 20mm forms the bottom layer. Green mix is loaded above the hearth layer. As soon as the raw materials reaches the ignition furnace, Top layer of green mix charge is ignited in the IGNITION FURNACE by burning of gases mainly CO gas . Air is drawn

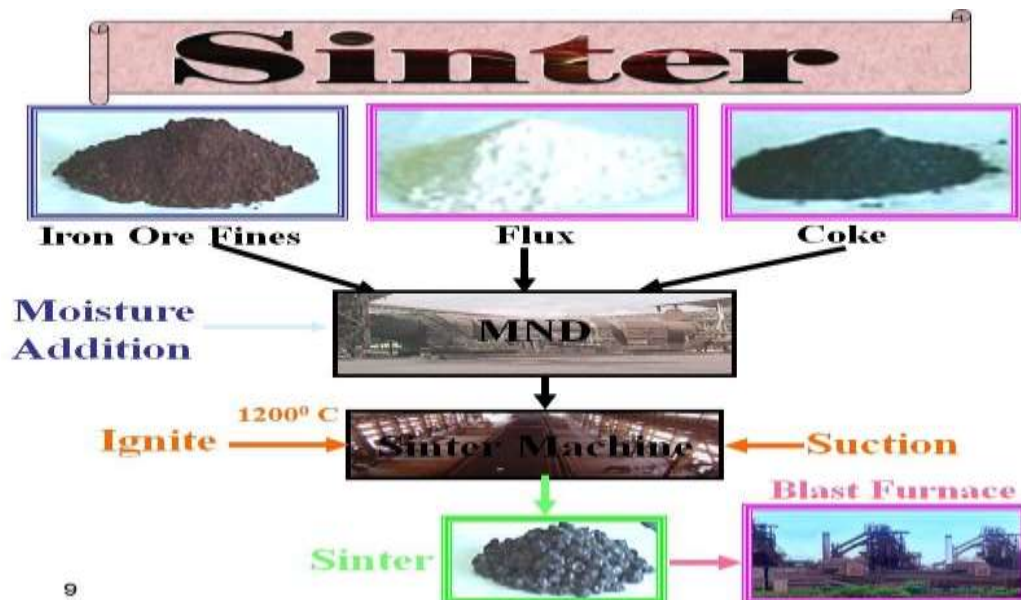
downwards through Exhausters or Waste Gas Fans. The heat from top layer is gradually transferred to subsequent bottom layers. . Due to burning of coke particles bonding take place between the grains and a strong & porous aggregate is formed known as “SINTER”. This sintering process is over when bottom layer coke fine burning is completed.

The sinter cake is then crushed, cooled, screened and dispatched to Blast furnace. The ideal size of sinter required in blast furnace is in between 5mm to 40mm. The - 5mm size sinters are screened & returned back to sinter bins.



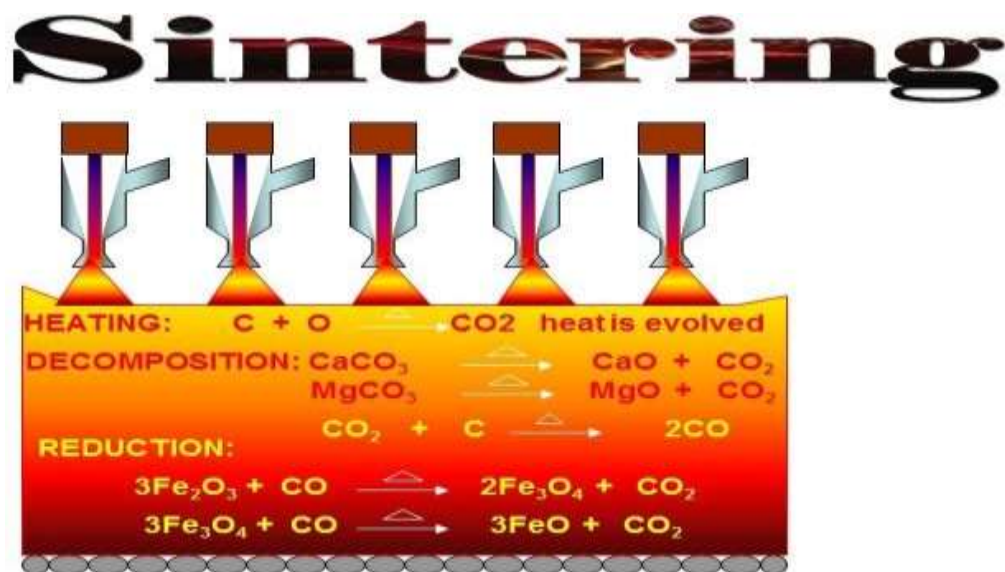
Fig1: Sinter Machine at SP3 , Machine 1

Sinter Making



Sintering of fines by the under grate suction method consists of the mixing of fines

with finely crushed coke as fuel and loading the mixture on the pallet grates. Ignition of the fuel proceeds on the surface of charge by a special ignition arrangement, called ignition furnace (where gaseous fuel is burnt to produce high temperature to ignite the fuel in sinter mix)



The gases used in ignition furnace are mainly coke oven gas or mixed gas. Mixed gas is combination of coke oven gas and blast furnace gas. Further the combustion is continued due to suction of air through the layers of the charge by means of Exhausters. Due to this, the process of combustion of fuel gradually moves downwards up to the grates.

From the scheme obtained in a few minutes after ignition, it is observed that the sintering process can be divided into six distinct zones:

- | | | |
|----|--------------------------------------|-------------------|
| 1. | Zone of Cold Sinter | (60 to 100 °C) |
| 2. | Zone of hot Sinter | (100 to 1000 °C) |
| 3. | Zone of intensive combustion of fuel | (1000 to 1350 °C) |
| 4. | Heating zone | (1000 to 700 °C) |
| 5. | Zone of Pre-heating of charge | (700 to 60 °C) |
| 6. | Zone of Re-condensation of moisture | (60 to 30 °C) |

In all the zones except the zone of combustion, the reactions taking place are purely thermal whereas in the zone of combustion reactions are thermal and chemical.

The maximum Temperature attained in the zone of combustion will be 1300-1350 °C. The vertical speed of movement of the zones depends on the vertical speed of sintering.

Heat from the zone of ready sinter is intensively transmitted to the sucked air. In the

zone of combustion of fuel hot air and preheated charge comes into contact with each other which with the burning fuel will result in the formation of high temperature. Maximum temperature will be developed in this zone and all the physical-chemical process takes place resulting in the formation of Sinter. In the zone of pre-heating the charge is intensively heated up due to transfer of heat from the sucked product of combustion. In the zone of re-condensation of moisture, the exhaust gases during cooling transfer excess moisture to the charge. Temperature of this zone sharply decreases and will not increase till all the moisture is driven off.

As the fuel in the zone of combustion is burnt away, Sinter, the height of which increases towards the grates, is formed above this zone from the red hot semi-fluid mass, forcing out subsequent zones. Disappearance of the zone of combustion means the end of sintering process.

The sinter cake is then crushed, cooled, screened and dispatched to Blast furnace. The ideal size of sinter required in blast furnace is in between +5mm to 40mm. The -5mm size are screened & returned back to sinter bin. (Called In plant return fines)

Following Approximate charge proportion will be required to make one ton of sinter (Wet basis):-

Ore fines	: 750-825kg
Coke	: 65-70 kg
Mill scale + fines	: 26 Kg
Lime stone	: 150-180 kg
B.O.F. Sludge	: 02kg
B.O.F. Slag	: 20Kg
Dolomite	: 30 - 40 kg
Burnt Lime	: 20 kg
BF Sinter return	: 100 kg
In plant sinter return	: 456 kg

Note- All above mentioned data varies in different plants under SAIL. Factors affecting sintering process:

1. Quality of Input raw materials

- a. Quality of Iron ore fines :
 - : +10 mm should be nil
 - : -1mm should be 30% maximum
 - : Alumina (Al_2O_3) 2.55% maximum
 - : Silica (SiO_2) 2.91% maximum

Increase in +10mm fraction will result in weak sinter & low productivity Increase in -1mm fraction will decrease bed permeability resulting in low productivity Increase in % of Alumina increases RDI (Reduction Degradation Index) resulting in generation of -5mm fraction & also resulting in chute jamming (Due to high Alumina in Base/Mix.

With increase of SiO₂ level in Iron ore fines, glassy phase in sinter increases and causes brittleness in sinter.

- b. Quality of Flux
 - : -3mm fraction should be 90% minimum (Crushing index)
 - : Less crushing index results in free lime, causing weak sinter
- c. Quality of Coke
 - : -3mm fraction should be 85% minimum (Crushing index of coke)
 - : +5mm fraction should be nil
 - : Increase in 5mm fraction decreases the productivity
 - : Increase in less than 0.5 mm particle size in coke causes increase in coke consumption during sintering

2. Moisture :

Moisture in the form of water is added in the base mix in Mixing/Nebulizing drum. Water acts as binder of base mix. Addition of water in base mix plays an important role in sinter bed permeability. Ideally 7 to 8% of total base mix of water is used. Higher % of water results in low permeability & less sintering speed. Less % of water results in less balling, hence less permeability, resulting in low productivity.

3. Ignition furnace temperature:

Ignition of sinter mix is carried out through ignition hearth where a temperature of 1150 to 1250 °C is maintained by burning gaseous fuel by the help optimum air/gas ratio.

32.5% of CO gas & 67.5% of BF gas is used to maintain calorific value 1900kcal/m³. Now a day Sintering Plant, Bhilai Steel Plant uses Coke Oven Gas of calorific value 4150 Kcal/Nm³.

Very Higher hearth temperature results in fusing of sinter at top layer. This reduces the bed permeability, hence low productivity. Low hearth temperature results in improper ignition. The sintering process will not be completed, hence -5mm fraction will increase, i.e. re-circulating load will increase.

Note- BF&CO gas mixing ratio and calorific value varies in different plants of SAIL

4. Coke rate :

Coke acts as a solid fuel in base mix in the sintering process. It is normally 3.5 to 6% of total charge. Higher coke rate will fuse the top layer, thereby decreasing the bed permeability. Sticker formation will increase. Low coke rate will result in incomplete sintering.

5. Machine speed :

The speed of sinter machine can be varied as per the condition of sintering process. BTP (Burnt Through Point) temperature decides the completion of sintering process. It is observed normally in second last wind box from discharge end side of sinter machine where the temperature reaches up to 400 °C (approximately). Higher machine speed, lower

BTP causes more –5mm generation, hence lower productivity. Lower m/c speed, higher BTP temperature causes low productivity.

Note: BTP: Exhaust gas temperature which indicates the completion of sintering process is called BTP. It is approximately around 400 degree centigrade.

Crushing, Cooling & Screening of Sinter

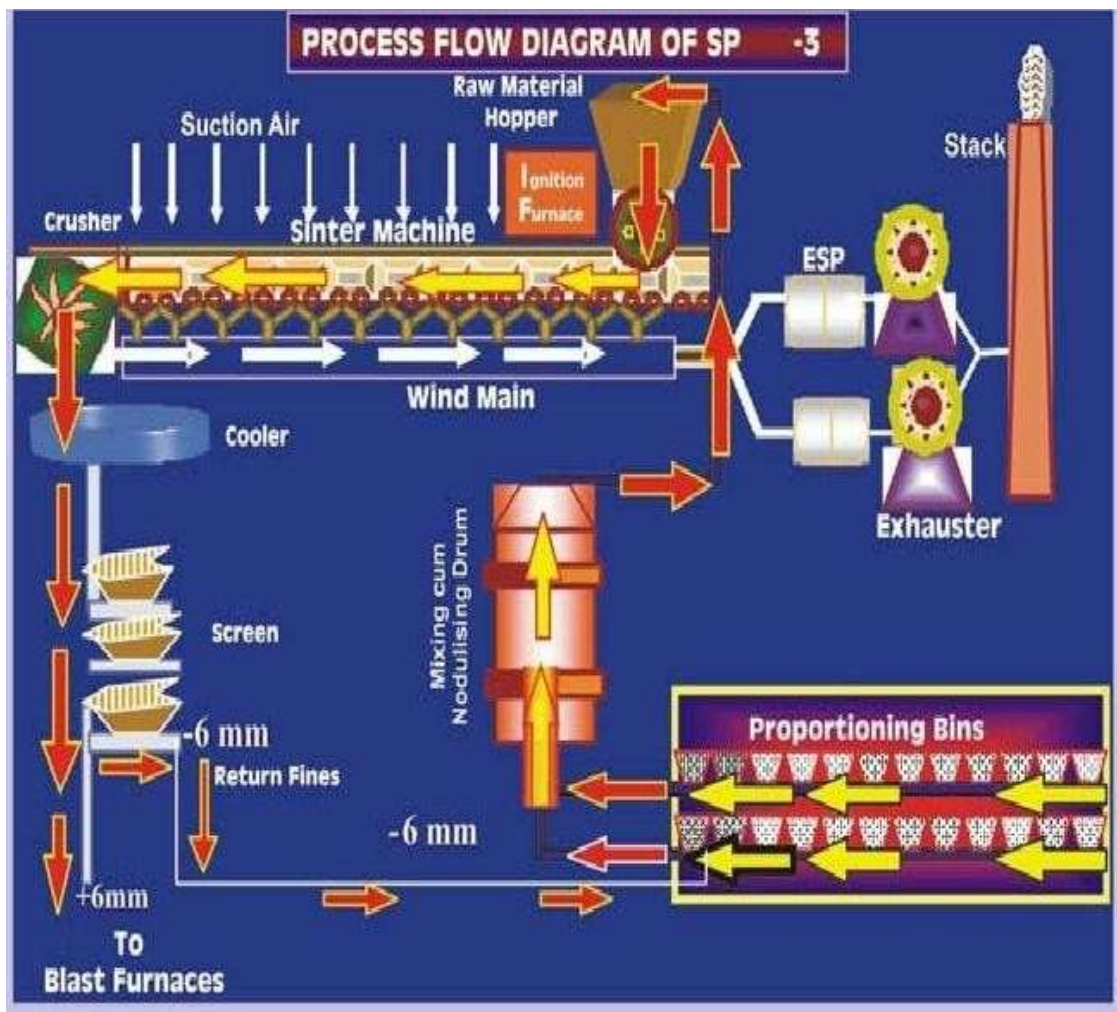
The finished Sinter cake is then crushed to the size of 100mm by using crushers. Cooling of finished crushed Sinter is then done on cooler by means of air blowers (forced draught fans), so that cooler discharge end temperature is about 60-80 degree centigrade. For effective cooling, bigger size of sinter should be on bottom portion & smaller size should be on the top.

Finally various fractions of Sinter are screened out. –5mm fraction of sinter, returns back to bunkers. 15 to 20mm fraction is also screened out to be used as hearth layer. Rest sizes goes to blast furnace. After screening, +10mm fractions should be 65% minimum and –5mm fraction should be 6% maximum as per requirement of blast furnace.

Advantages of using Sinter

1. To utilize the ore fines generated at mines to transform to an acceptable feed in blast furnace
2. To utilize economically all the metallurgical wastes like Mill scale, L.D slag, B.O.F slurry, Flue dust, Ferro scrap etc.
3. To utilize the coke breeze generated in coke screening at coke ovens as fuel, otherwise has no metallurgical use
4. As the calcination of flux takes place in sinter strand, super-fluxing saves much more coke in the furnace.
5. Increase of sinter percentage in Blast Furnace burden, increases the permeability, hence reduction and heating rate of burden increases, so the productivity also increases. Coke rate is also reduced in Blast furnace.
6. Minimal fraction of total mass of impurities, Viz. sulphur, phosphorous, zinc, alkali is reduced .
7. Improved quality of hot metal.
8. The softening temp. of sinter is higher and melting zone is narrow. This increases the volume of granular zone and shrinks the width of cohesive zone consequently, the driving rate of BF become better.

PROCESS FLOW DIAGRAM OF SINTER PLANT



3.3 Quality Parameters of Sinter (Subject to Requirement of BF)

	Chemical composition		Physical composition	
1.	FeO %	8.0 to 11.0	Sinter size	5mm to 40mm
2.	MgO %	2.6 to 3.0	Mean size	18mm to 21mm
3.	Available lime (CaO-SiO ₂)%	3.4 to 6	DTI	70% MIN
4.	As per BF Requirement		RDI	30% MAX
5.	SiO ₂ %	4.8 to 5.2	+ 10 mm	65 % min.
6.	Al ₂ O ₃ %	3.0	+40 mm	9 % max.
7.	Basicity.	1.6 to 2.1	- 5 mm	6% max.

Note- Quality parameters of sinter varies in different plants under SAIL.

Quality parameter definitions:

Tumbler index (DTI): The cold strength of sinter is determined by the tumbler test, and depends on the strength of each individual ore component, the strength of the bonding matrix components and the ore composition. This test determines the size reduction due to impact and abrasion of the sinters during their handling, transportation, and in the blast furnace process. Studies of the fracture strength of several mineral phases have allowed the following order to be established: primary (or residual) hematite > secondary hematite > magnetite > ferrites. Cold mechanical strength is directly related with the tendency for fines to form during transportation and handling between the sinter machine and the blast furnace throat.

Reduction Degradation Index (RDI)

Sinter degradation during reduction at low temperature is more usually determined by the RDI static test, which is carried out at 550 °C. Low values are desirable for this index. The RDI is a very important parameter that is used as a reference in all sintering work and serves to predict the sinter's degradation behavior in the lower part of the blast furnace stack.

Some critical terms/parameters used/monitored in sinter plant:

Coke crushing index	Percentage presence of –3mm fraction of coke in any sample is termed as coke crushing index. For better sintering process coke crushing index should be more than 85%
Flux crushing index	Percentage presence of –3mm fraction of flux in any sample is termed as Flux crushing index. For better sintering process Flux crushing index should be more than 90%
Burn Through Point (BTP)	Burn through point temperature indicates the completion of sintering process. It is normally around 400 degree Celsius and is normally found in second last of wind box from discharge end of sinter machine.

3.4 Main Areas & Equipment

<u>Main Areas</u>	<u>Equipments</u>	<u>Functions</u>
Sinter making & Cooling bldg.	Balling drums Sinter pallets Screens Crushers Coolers	To mix & pelletize Sintering takes on it Screens out diff. sizes Crushes sinter cake Cools/ Normalize sinter
Exhausters	High capacity fans Battery cyclones ESP	To suck air below grates To clean Exhaust air To clean Exhaust air

Proportioning Bins	Electronic feeders Conveyors Bunkers	For adjusting feeding Transport charge mix. Store raw materials
Coke & Flux Crushers	Roll crushers Rod Mills Hammer crushers Grab cranes	For crushing coke For crushing coke For crushing Fluxes For lifting coke

Techno Economics

1. Specific Productivity : Sinter produced per square meter per hour
2. Specific Heat consumption : Gas consumed per ton of sinter
3. Specific Power consumption : Power consumed per ton of sinter
4. Specific Coke consumption : Coke consumed per ton of sinter
5. Specific Flux consumption : Flux consumed per ton of sinter

In order to produce sinter at less cost, specific productivity of sinter should be as high as possible & all other four parameters should be as low as possible keeping quality parameters under consideration.

Advantages of Sintering

1. Better use of the huge quantity of iron ore fines generated at mines.
2. Gainful use of various metallurgical wastes like flue dust, mill scale, lime dust, sludge, etc.
3. Use of super fluxed sinter eliminates raw flux from the blast furnace burden. This leads to considerable coke saving and productivity improvement in blast furnaces.
4. Due to the higher reducibility of super fluxed sinter, direct reduction of iron oxide is enhanced, which contributes to further coke saving.
5. The softening temperature of sinter is higher and the softening melting zone is narrower. This increases the volume of granular zone and shrinks the width of the cohesive zone. Consequently, the driving rate of the blast furnace improves.
6. Hot metal quality (from the SMS point of view) improves due to lower silicon content and higher hot metal temperature. A higher hot metal temperature contributes to better sulphur removal from the hot metal.
7. Material handling in the charging section of the blast furnace is reduced, and fewer logistics are needed.
8. Blast furnace operation is more reliable and efficient

3.5 Safety hazards at Sinter plant

1. Dust pollution : As lot of finer particles are used in sintering, there causing lots of dust pollution. Efficient running of ventilation is must. Use of dust mask is essential. Chimney Stack Emission is $50\text{mg}/\text{nm}^3$. Fugitive Emission(ambient) is $2\text{mg}/\text{nm}^3$
2. Gas safety : Gases (usually Mixed gas & Coke oven gas) are used for igniting charge mix, It is very important to follow all the protocols for gas safety. Use of gas mask and Carbon mono oxide (CO) gas monitor while working on gas line is must.

3. Noise pollution: Tremendous amount of air is sucked through exhausterfans. Slight leakages anywhere in suction line or exhauster results in high level of noise. Air compressor, chiller unit, hammer crusher, coke crusher are also high noise generating areas in Sinter plant. Use of Ear plug is essential.

3.6 ISO 45001:2018 (Occupational Health and Safety Management System):

OH&SMS provides a formalized structure for ensuring that hazards are identified, their impact on staff assessed and appropriate controls put in place to minimize the effect. It further assists a company in being legally compliant, ensuring appropriate communication and consultation with staff, ensuring staff competency and having arrangements in place to deal with foreseeable emergencies. It is not concerned with the safety of the product or its end user.

It is compatible with the established ISO 9001(Quality) and ISO 14001 (Environmental) management system standards. This helps to facilitate the integration of the quality, environmental and occupational health and safety management systems within the organization.

Impacts of fully implemented OH&SMS are:

- a) Risks and losses will be reduced and/or eliminated
- b) Reduced accidents, incidents and costs
- c) Reliable operations
- d) Compliance to rules, legislation, company standards and practices
- e) A systematic and efficient approach to health and safety at work
- f) Positive company image and reputation